

LLM TRAINING PIPELINE

How generative AI models are built — from raw data to deployed product

1 THE TRAINING PROCESS

1

Data Collection & Curation

Billions of web pages, books, code repositories, and scientific papers are gathered, filtered for quality, deduplicated, and converted into tokens.

Trillions of tokensWeb · Books · Code · Papers

2

Pre-Training

The model learns by predicting the next token across the entire dataset. This is the longest and most expensive phase — run on massive GPU clusters 24/7.

10,000–50,000 GPUsWeeks to months · ~\$10M–\$100M+

3

Fine-Tuning (SFT)

The pre-trained model is further trained on curated instruction-response examples, teaching it to follow directions and be genuinely helpful.

Thousands–millions of examplesDays to weeks

4

RLHF / Constitutional AI

Human raters rank model outputs; a reward model learns from these preferences. The policy is then optimized to score higher — shaping safety and quality.

Human preference dataReward model · PPO / RLAI

5

Evaluation & Safety Testing

Rigorous testing on standardized benchmarks, adversarial red-teaming, and bias audits confirm the model is ready for release.

MMLU · HumanEval · TruthfulQARed-teaming · Bias audits

6

Deployment

The model is served through an API or consumer product. Optimizations like quantization and KV caching reduce inference cost at scale.

API / consumer productQuantization · KV cache

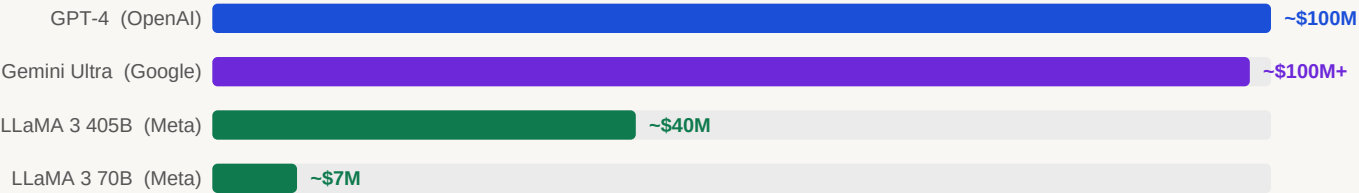
2 RESOURCES REQUIRED

DATASETS	COMPUTE	ENERGY	TIME	COST
Trillions of tokens Web, books, code, papers	10k–50k GPUs/TPUs Running 24/7 for weeks	100s–1,000s MWh Per frontier training run	1–6 months Pre-training phase only	\$10M–\$100M+ Compute costs, frontier models

3 REAL MODELS — TRAINING RESOURCES & COSTS

GPT-4	Claude 3 / 4	LLaMA 3 (405B)	Gemini Ultra
OpenAI · 2023 ~\$100M est.	Anthropic · 2024–25 Undisclosed	Meta · 2024 ~\$30–50M est.	Google · 2023 ~\$100M+ est.
~1–10T tokens ~25,000 A100 GPUs - RLHF	- Web + curated - Undisclosed GPUs - Constitutional AI	~15T tokens ~24,000 H100 GPUs - SFT + RLHF	- Multimodal data - TPU v4/v5 pods - RLHF

4 TRAINING COST COMPARISON (ESTIMATED)



* All cost figures are third-party estimates. Actual costs undisclosed by most labs. Figures reflect compute only.